

Wildfire Smoke & PM2.5 Forecasting¹

CS2: Transformer-Based Air Quality Prediction

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Why PM2.5 Forecasting Matters

Motivation

The asthma rate in the BC area has nearly doubled, rising from 5.25% to 9.88%.

Primary Stakeholder

Environmental Health Director — responsible for timely, defensible air-quality advisories used across provincial partners.

CS1 Limitations and CS2 Objectives

CS1 Limitations

- Relied on historical PM2.5 patterns; using too few features risks over-reliance on a single factor.
- Lacked short- and long-term memory, which is problematic for time-series data.
- The model was too conservative, causing it to miss rapid, non-linear spikes (notably from wildfire smoke).

CS2 Objectives

- Expand the feature set to capture environmental complexity.
- Incorporate memory into the model to better handle time-series dynamics.
- Improve the model's sensitivity so it is less conservative during extreme events.

Two data sources (2022–2025):

- **PM2.5** — CS1 pipeline (BC AQMS, zone-median, hourly)
- **Wildfire** — CWFIS satellite hotspots

Key design decisions:

- Zone **polygon boundary** distance, not centroid
- **300 km** regional radius — cross-zone smoke transport
- IDW weighting ($1/d^2$) — closer fires weighted more
- Non-fire hours **zero-filled**, not missing

Wildfire signals:

- fire_count, frp_sum (local & regional)
- hfi_weighted, fwi_mean

Data Structure and Modeling Assumptions

Data Structure:

- 70,128 rows $\times$ 11 features, 2 zones, hourly UTC

Data Augmentation:

- **PCHIP interpolation** — upsample hourly $\rightarrow$ 30-min
- Doubles training sequences; smoother PM2.5 transitions
- *Assumption: more data improves model generalization*

Data Assumptions:

- ① **Wildfire drives PM2.5** — spikes in BC air quality are primarily caused by wildfire smoke
- ② **300 km influence radius** — fires within 300 km of a zone boundary meaningfully affect local PM2.5
- ③ **Zone independence** — Lower Fraser Valley and Southern Interior modeled separately; no cross-zone features

Why Transformer? Model Selection and Trade-offs

From Baseline GAM to Transformer

Model	Improves	Still limited by
GAM (baseline)	Nonlinear structure	Manual features, weak sequence modeling
LightGBM	Identify useful predictors for nonlinear tabular learning	Limited time-series handling
LSTM + Autoencoder	Sequence learning + feature compression	Slow sequential training

Therefore, we chose Transformer: Captures temporal dependence more flexibly and supports parallel training.

Why Transformer

- **Sequence-aware** models historical PM2.5 patterns directly
- **Long-range dependence** can connect distant time steps more effectively
- **Flexible representation** captures nonlinear temporal structure
- **Parallel training** more scalable than recurrent models

Trade-offs

- **Higher compute cost** larger and more complex than GAM
- **More tuning required** architecture and optimization settings matter
- **Lower interpretability** harder to explain than smooth GAM effects
- **Overfitting risk** requires careful validation and regularization

Model Architecture and Design Choice

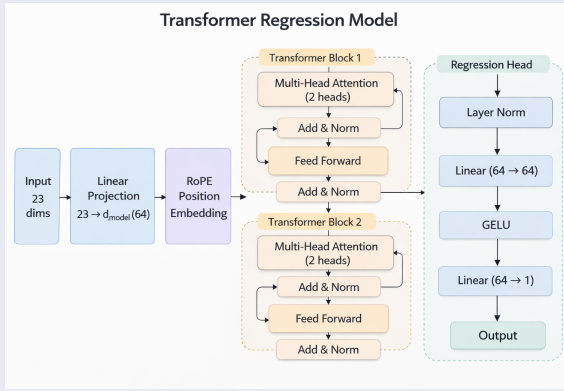
Transformer Model Structure

Architecture

- Transformer encoder
- Window size: **96 observations**
- $d_{\text{model}} = 64$
- 2 attention heads
- 2 Transformer layers
- dropout = 0.1

Total parameters size: 0.106M

Total parameters intentionally **kept small** to reduce overfitting.



Why Not More Complex Alternatives?

We experimented with **more complex architectures** inspired by LLaMA:

- SwiGLU feed-forward layers
- Larger hidden dimensions
- Deeper Transformer blocks

However:

- Dataset size relatively small: 26,245 data entries (12.05 MB) per zone
- Models trained **separately per zone**
- Complex variants did **not improve stability or performance**

Final decision:

Use a simpler Transformer block with strong regularization.

Training Setup

Optimizer: **AdamW**

Loss: **MSELoss**

Training parameters:

- Batch size: 256
- Epochs: 10
- Learning rate: $5e-4 \rightarrow 5e-6$ (cosine decay)
- Weight decay: 0.05

Feature and Window Experiments

Feature set includes:

- PM2.5 lags and rolling statistics
- Fire activity indicators (FRP, fire counts)
- Temporal encoding (hour, weekday, month)

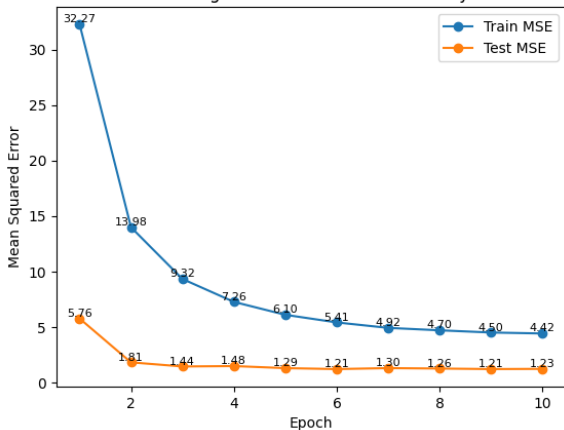
Window length experiment:

12, 24, 48, 72, **96**, 168 observations

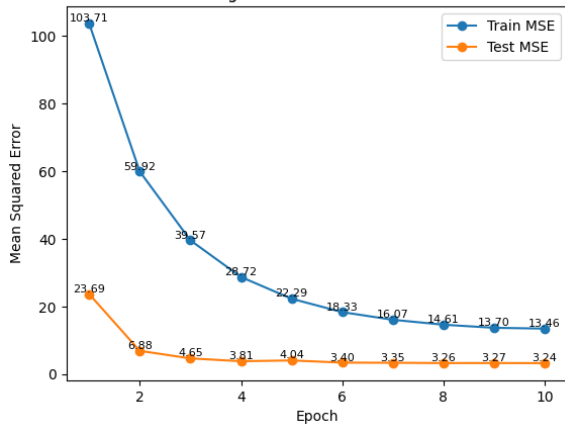
Best performance observed at **96 observations**.

Training Dynamics

Training Curve — Lower Fraser Valley



Training Curve — Southern Interior



Training monitored using **train and test loss curves** across 10 epochs.

Leakage Prevention

Several safeguards were applied:

- StandardScaler **fit only on training data**
- Test data transformed using training statistics
- Sequence windows constructed **after dataset split**
- No window crosses train–test boundaries

Zone-Specific Modeling

Models trained **independently** for each region:

- Lower Fraser Valley
- Southern Interior

This avoids mixing **regional wildfire patterns**.

Model predictions follow **PM2.5 trends in the unseen test year**, demonstrating generalization to new wildfire seasons.

Main Performance Comparison

Zone	Model	Precision@25	Recall@25	F1@25
Lower Fraser Valley	Baseline	0.500	0.538	0.518
	Transformer	0.538	0.538	0.538
Southern Interior	Baseline	0.945	0.896	0.920
	Transformer	0.945	0.888	0.916

- Metrics are evaluated at a **PM2.5 threshold of 25 $\mu\text{g}/\text{m}^3$** , where values at or above 25 are treated as high-pollution events.
- Model performance differed sharply by air zone.
- Southern Interior performed strongly: Precision 0.945, Recall 0.888, F1 0.916.
- Lower Fraser Valley performed much worse: all key metrics were 0.538.
- Overall, the model appears reliable in the Southern Interior but much less dependable in the Lower Fraser Valley.

Diagnostics and Air-Zone Comparison

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	Transformer	0.538	0.538	0.538
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- Diagnostics show that performance is highly region-dependent.
- The Southern Interior appears to have clearer or more stable event patterns, which the model captured well.
- The Lower Fraser Valley appears to be a more difficult forecasting environment, leading to weaker warning performance.
- This comparison shows why zone-level evaluation matters: overall results alone would hide important regional differences.

Limitations, Uncertainty, and Final Takeaways

- Results should be interpreted with caution because performance is not consistent across air zones.
- The weaker Lower Fraser Valley results indicate greater uncertainty and a higher risk of unreliable warnings.
- The approach shows clear promise, especially for the Southern Interior.
- Main takeaway: the model has value, but further refinement and zone-specific tuning are needed before broader operational use.

Section 1

Appendix

Canadian Wildfire Information System (CWFIS):

- Satellite hotspot detections from **MODIS** and **VIIRS** sensors
- Data provided in **UTC** timezone
- Matched to PM2.5 hourly records after timezone alignment (PST $\rightarrow$ UTC +8h)

Wildfire Signal Definitions

Signal	Description
fire_count	Number of active fire hotspots within radius
frp_sum	Total Fire Radiative Power (MW) within radius
hfi_weighted	IDW-weighted Head Fire Intensity
fwi_mean	Mean Fire Weather Index within radius

Local signals use the zone's **polygon boundary**; regional signals use a **300 km radius**.